



OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

4. Evolutionary Process Model:- Spiral approach

Archana A

Department of Computer Applications

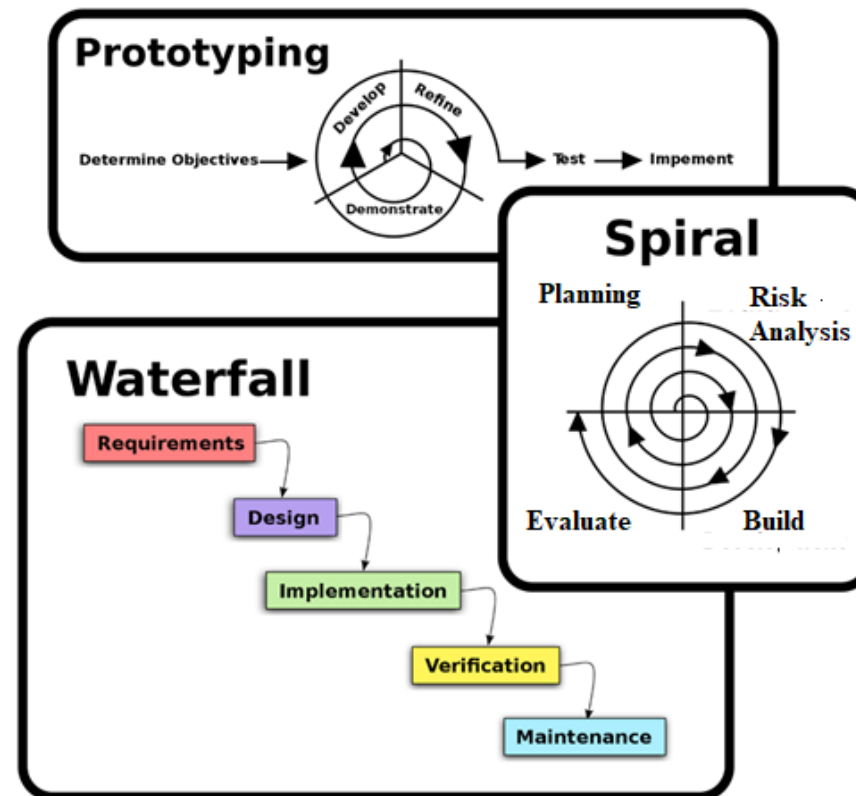
(ii) Spiral Model

Main characteristics

- Iterative.
- Prototype-oriented.
- Starts with **planning** and ends with **customer evaluation**.
- Low risk.

- The spiral model is **similar to the incremental model**, with more **emphasis** placed on **risk analysis**.
- The spiral model can be **adapted to apply throughout the entire life cycle** of an application, from concept development to maintenance.
- Spiral **models uses prototyping** as a **risk reduction mechanism** but, more important, enables the developer to apply the prototyping approach at each stage in the evolution of the product.

- It maintains the **systematic stepwise approach** suggested by the classic life cycle (i.e, waterfall) but **also incorporates** it into an **iterative framework activity** (prototype approach).

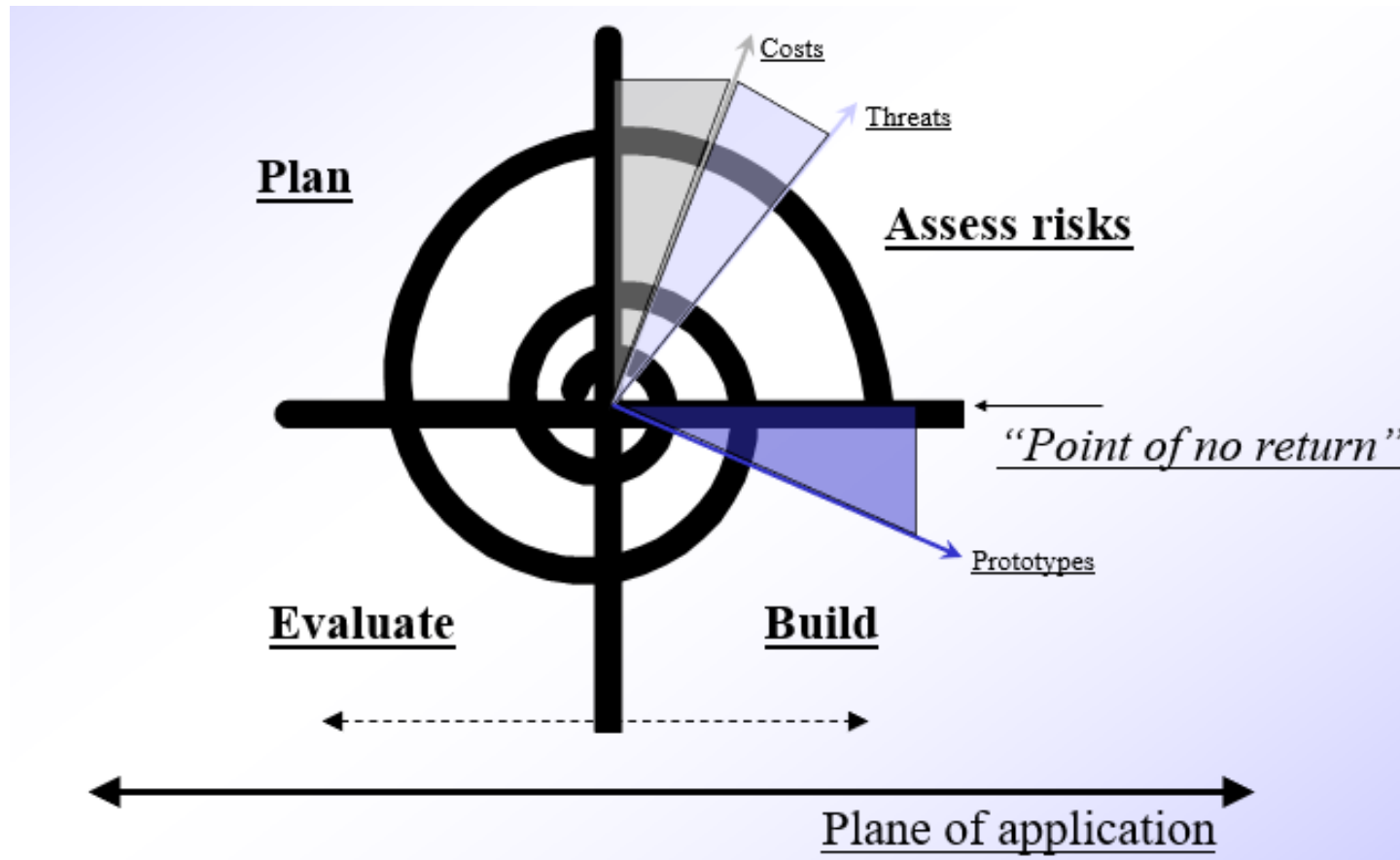


Spiral areas are:

- **Planning**
 - Getting requirements
 - Project planning (based on initial requirements.)
 - Project planning (based on customer evaluation.)
- **Risk analysis**
 - Cost/Benefit and threats/opportunities analysis
 - Based on initial requirements and later on customer feedback
- **Engineering**
 - Preview it
 - Do it
- **Customer evaluation**

OBJECT ORIENTED SOFTWARE ENGINEERING

4. Evolutionary Process Model - (ii) Spiral Model...



- Each phase of Spiral Model is divided into **four quadrants** as (*shown in the previous slide*). The functions of these four quadrants are:

1. **Planning:-** *Objectives determination and identify alternative solutions.* Requirements are gathered from the customers and the objectives are **identified, elaborated** and **analyzed** at the start of every phase. Then **alternative solutions** possible for the phase are **proposed** in this quadrant.

2. Risk Analysis:- *Identify and resolve Risks.* During the second quadrant all the possible solutions are evaluated to select the best possible solution.

- Then the **risks associated** with that **solution is identified** and the risks are **resolved** using the best possible strategy.
- At the end of this quadrant, Prototype is built for the best possible solution.

3. Engineering :- *Develop next version of the Product.* During the third quadrant, the **identified features are developed** and **verified through testing**. At the end of the third quadrant, the **next version of the software is available**.

4. Customer Evaluation:- *Review and plan for the next Phase.* In the fourth quadrant, the **Customers evaluate** the so far developed version of the software. In the end, planning for the next phase is started.

1. **Risk Handling:** High amount of **risk analysis** hence, avoidance of Risk is enhanced.
2. **Good for large** and mission-critical (complex) projects.
3. Strong approval and documentation control.
4. **Flexibility in Requirements:** Additional **Functionality can be added** at a later date.
5. **Customer satisfaction:** **Software is produced early** in the software life cycle.

1. **Expensive:** Can be a **costly model** to use.
2. To perform Risk analysis, it requires **highly specific expertise**.
3. **Doesn't work** well for **smaller** projects.
4. **Too much dependable on Risk Analysis:** Project's **success** is highly **dependent on the risk analysis phase**.
5. **Difficulty in time management:** As the number of phases is unknown at the start of the project, so time estimation is very difficult



THANK YOU

Ms. Archana A

Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392